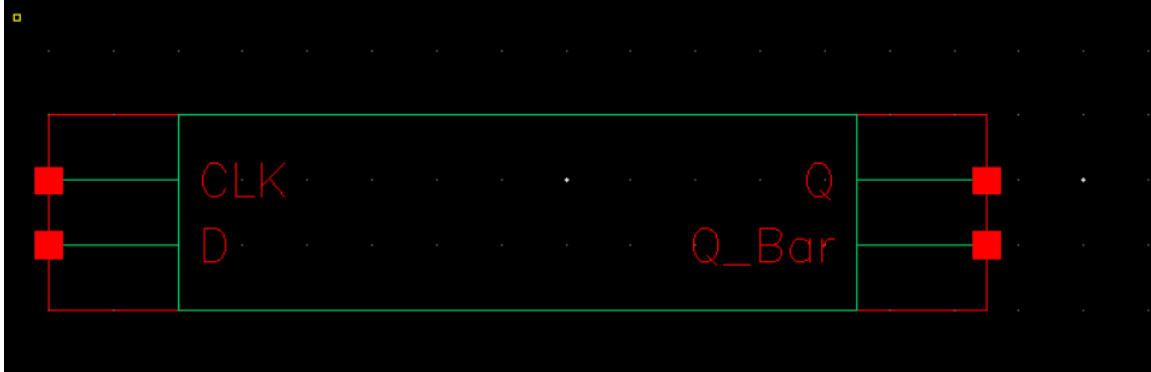
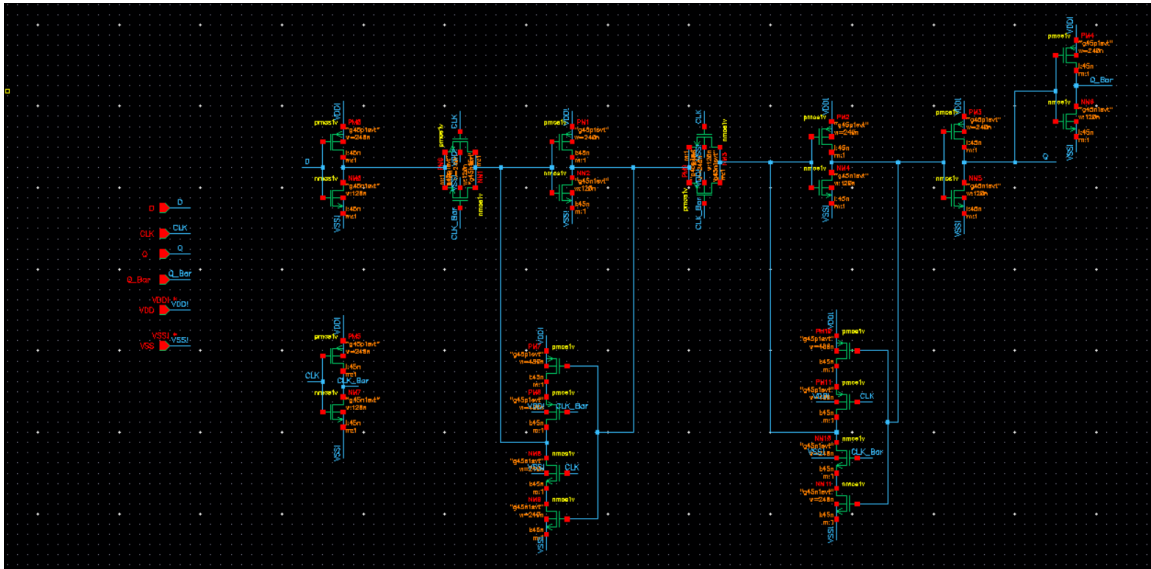


Library Name:		cgm	
Cell Name:		DFFX1	
Function/Truth Table:			
CLK	D	Q	Comment
↑ (rising)	0	0	Q follows D on rising edge
↑ (rising)	1	1	Q follows D on rising edge
0 or 1 (not ↑)	X	Q(No Change)	No rising edge → Q holds its value
Set Up Time D=1			
Pre-layout		70 ps	
Post-layout		100 ps	
Hold Time D=1 (if not negative)			
Pre-layout		-50 ps	
Post-layout		-70 ps	
Clock to Q-Delay			
Pre-layout		1.70397ns	
Post-layout		1.73977ns	
Layout Height: 1.71nm			
Layout Width: 8.4nm			
Layout Area: 14.364 nm²			

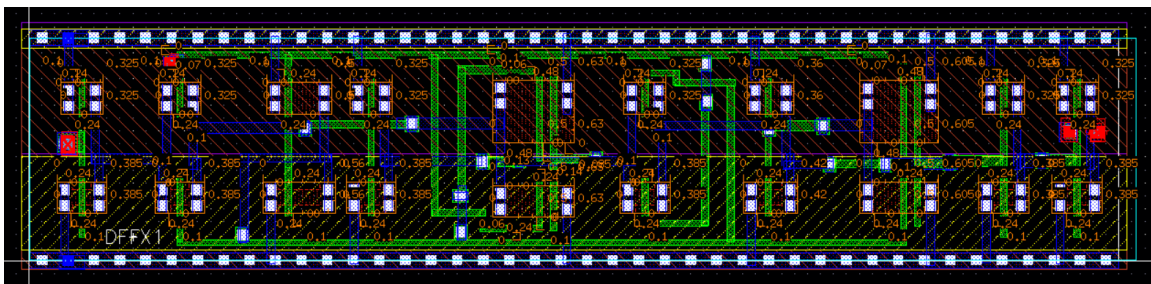
Symbol with Port Names:



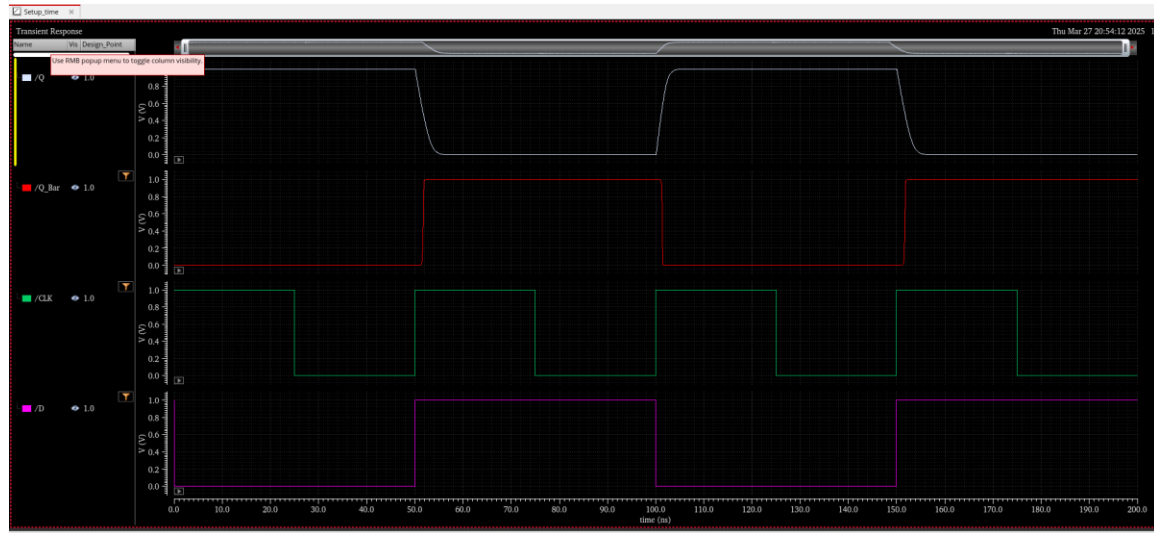
Schematic:



Layout:



Functional Simulation Waveforms



Comments/Notes:

In both pre-layout and post-layout simulations, the flip-flop exhibits negative hold time behavior. During pre-layout simulation, failure was observed when D changed between 49.930 ns and 49.950 ns, but Q correctly latched the data when D changed at 49.960 ns, resulting in a hold time of -50 ps. Similarly, in the post-layout simulation, failure occurred when D changed between 49.900 ns and 49.930 ns, but worked correctly at 49.890 ns, confirming a hold time of -70 ps.

This behavior indicates that the flip-flop can reliably latch data even when the input D changes slightly before the rising edge of the clock. This is due to internal delays where the data path is slower than the clock control path, creating a small timing margin that allows early transitions on D without corrupting the output. In essence, negative hold time means the flip-flop is tolerant to slight timing violations immediately after setup, and can capture data reliably even when D changes just before the clock.